SPARK CHAPTER

Islamabad, Pakistan

The Aristotelian Classroom

A Six-Session Experiment in Voluntary Learning, Free Enquiry, and the Recovery of Curiosity

SPARK Summer Camp 2026 — Case Study and Findings Report

35 registered participants · 6 sessions · 67 feedback submissions · July–August 2026

Prepared for partner schools, prospective partners, and general readership

All participant data anonymised. Quotations reproduced verbatim from voluntary feedback forms.

Contents

Abstract

Between July and August 2026, SPARK Chapter ran a free six-session summer camp for 35 school and university students in Islamabad. The camp was built to test a specific proposition: that the psychological burden commonly associated with formal schooling is not produced by intellectual difficulty, but by the evaluative apparatus wrapped around it — attendance registers, graded homework, and an authority figure whose role is to render a verdict on whether a student is capable.

We removed that apparatus entirely. There was no attendance requirement, no grading, no compulsory homework, and no examination. Participants were told what to study before each session, and were never checked. Every session ran in two blocks: a technical block covering Python, data handling, machine learning and web development, and an open discussion block with no fixed syllabus, in which the topic was whatever the room was curious about — abiogenesis, epistemology, theology, mythology, cinematography, history, physics, art. Facilitators moderated and explained on request; they did not lecture unless asked, and did not evaluate at any point.

The results describe a clear and repeatable pattern. Participation collapsed to roughly 50% by the second session as students attempted to interpret a room that behaved unlike any classroom they had encountered, then recovered to approximately 86% by the fourth session and held there to the end. The same J-shaped curve appears independently in three measures: attendance, voluntary feedback submission, and willingness to speak. We interpret the trough not as failure but as the measurable cost of adaptation to non-evaluative learning — approximately two sessions and half the room.

On outcomes, the divergence between the two tracks was sharp and instructive. The discussion block was rated the single most valuable component by nearly every participant and produced the outcome the camp was actually designed for: a documented, unprompted increase in self-generated questions and independent reading. The technical block produced high engagement and low measured skill transfer, which we attribute to structural factors — extreme heterogeneity of prior preparation within a single cohort, and a weekly cadence that provides insufficient rehearsal for cumulative skills — rather than to the absence of compulsion.

The central finding is a mechanism rather than a score. Across multiple independent cases, participants who encountered material they could not understand did not disengage and did not conclude they were incapable. They attributed the difficulty to method, preparation or circumstance, and stated an intention to continue. We propose that this re-attribution is the operative effect of removing evaluation: difficulty stops functioning as a verdict, and therefore stops producing avoidance. This is a narrower claim than the abolition of academic rigour, and a considerably more actionable one for schools.

1. About SPARK Chapter

SPARK Chapter is a student-led, non-profit youth education initiative based in Islamabad, operating in the vicinity of the NUST H-12 campus. It was founded on the conviction that the most significant deficit in Pakistani secondary education is not access to information — which is now effectively free — but the near-total absence of environments in which a young person may think aloud, be wrong in public, and follow a question wherever it leads without being scored for it.

SPARK is run by volunteers and delivers its programmes at no cost to participants. It takes no fees, sets no entry examinations, and imposes no academic prerequisites. Its facilitators are practitioners rather than career educators: working engineers, founders and university students who teach because they are still close enough to the experience of learning to remember what it costs.

1.1 Position

SPARK does not attempt to replace or duplicate school. Schools are structurally obliged to certify. Certification requires measurement, measurement requires standardisation, and standardisation requires that every student be moved through the same material at the same rate and judged against the same threshold. That machinery is not a defect of any individual institution — it is the price of issuing a credential society can trust.

Our position is that this machinery has a side effect which is rarely examined because it is rarely separable: it teaches students that not-knowing is a verdict on their capacity. SPARK exists to occupy the space that formal schooling structurally cannot occupy, and to run programmes in which the cost of admitting ignorance is zero.

1.2 Programmes

SPARK's activities include workshops, in-school sessions delivered in partnership with local institutions, strategic and simulation-based learning games, and multi-week camps of the kind documented in this report. This document concerns the Summer Camp 2026 cohort, which was the organisation's most instrumented programme to date and the first designed explicitly as an experiment with a stated hypothesis and a pre-committed measurement approach.

2. The Question

The camp was designed around a question we had not seen tested in our own context, and which is normally argued rather than measured. It sits between two well-known positions separated by twenty-three centuries.

2.1 The Aristotelian premise

Aristotle's Lyceum has come down to us as the peripatetic school — the walking school — in which enquiry proceeded by conversation between people of unequal knowledge but equal standing to speak. Instruction was not delivered on a fixed syllabus to a seated and silent audience; a question was raised, it was pursued, and it led where it led. The teacher's contribution was not the transmission of a curriculum but the sharpening of an argument already in motion.

Two features of this model are frequently misread. First, it was not unstructured — the Lyceum had norms of conduct and argument, and expected participants to defend claims. Second, it was not egalitarian in knowledge — Aristotle plainly knew more than his interlocutors and said so. What it lacked was positional authority: no one in the room held the power to certify whether another person was capable. The asymmetry was in knowledge, not in judgement.

This is the specific property we set out to reproduce. Our shorthand for it in planning was organised freedom of thought and expression: a room with real norms, real expertise available on demand, and no verdict.

2.2 The Sagan problem

Carl Sagan observed that young children ask questions of remarkable ambition and frequency, and that by high school this has largely stopped. His conclusion was that something happens in between, and that it is worth identifying, because the loss is not natural maturation but damage.

> *Children ask enormously important questions. But by the time they get to high school they have almost stopped asking. Something terrible has happened between kindergarten and high school.*
> — Carl Sagan, paraphrased from The Demon-Haunted World

We take a specific view of what happens in between, and it is not that children lose interest. A five-year-old asking why the sky is blue risks nothing. A sixteen-year-old asking the same question in a graded classroom risks being marked, by peers and by the institution, as someone who does not already know. In an environment where every act of participation is potentially assessed, the rational strategy is silence. Curiosity is not extinguished — it is priced out.

If that account is correct, then curiosity should be recoverable. It requires no motivational intervention and no novel pedagogy. It requires only that the price be removed and that enough time be allowed for students to believe the price really has been removed.

2.3 Hypothesis

> **Primary hypothesis**
>
> The psychological burden of formal education is produced not by intellectual difficulty but by evaluation. In an environment where difficulty carries no verdict on the learner's capacity, students will voluntarily choose difficult material, will persist through failure, and will resume asking questions of their own.

Three subsidiary questions followed from this:

1. Under genuinely voluntary conditions — no attendance, no grading, no consequence for leaving — will students attend a free programme, and will they return after difficulty?
2. Will a discussion environment with moderation but no lecturing produce substantive intellectual engagement, or will it collapse into unfocused conversation?
3. Given free choice, what structure will students request for themselves? This we regarded as the most informative question available, because a constraint chosen is not a burden, and what learners ask for when nothing is imposed is a direct measurement of what they actually need.

3. Programme Design

3.1 Structure

The camp comprised six sessions delivered between 22 July and 13 August 2026, running principally on a weekly Wednesday cadence with the final two weeks doubled to consecutive days. The programme had been publicly announced approximately two weeks before it began and was then delayed — a circumstance which, as recorded in Section 5.4, proved unexpectedly informative.

Session	Date	Technical block	Discussion block (representative)
Day 1	22 Jul 2026	Orientation; first exposure to code	Goals, passions, work and earning; politics; industry practice
Day 2	29 Jul 2026	Mathematical foundations; mechanics	History; ideology; personal direction
Day 3	5 Aug 2026	Calculus; NumPy and arrays	Open-floor enquiry; participant presentations
Day 4	6 Aug 2026	Libraries; Matplotlib; data handling	Gender and feminism; social structure; physics
Day 5	12 Aug 2026	Model training and prediction	Participant presentations; theology; mythology
Day 6	13 Aug 2026	Consolidation; applied direction	Passion and interest; closing synthesis

Table 1. Session calendar. Discussion topics are indicative only; no discussion syllabus existed in advance.

Each session ran two blocks. The technical block covered Python, mathematical foundations, data handling with NumPy, Pandas and Matplotlib, introductory machine learning, and elements of web development. The discussion block had no syllabus at all. Over six sessions it ranged across abiogenesis, theology, epistemology, history, geography, literature, art, philosophy, cinematography, physics, and the formation of mythologies including vampire folklore — in every case because a participant raised it, and in most cases because it emerged from something else already under discussion.

3.2 Rules of the room

The programme operated under a small number of explicit commitments, stated to participants and adhered to without exception:

- No attendance requirement. Participants came when they wished and left when they wished. Absence required no explanation and carried no consequence.
- No grading, marks, ranking or assessment of any kind, at any point.
- No compulsory homework. One optional task was set across the six sessions. Otherwise participants received a short message before each session indicating what they might read or attempt — framed explicitly so that they would arrive with questions, not so that they would arrive having complied.
- No lecturing in the discussion block. Facilitators moderated, de-escalated, and explained concepts when asked. They did not set topics and did not deliver prepared material.
- No mockery. Ideas were explored rather than dismissed, including ideas the facilitators considered mistaken. This norm applied to facilitators first.
- Continuous access. Facilitators remained available to participants outside sessions through a shared community channel, during the camp and after its conclusion.

On the facilitator relationship

Facilitators did not present as teachers and stated this explicitly in the room. The relationship framed was closer to that of an older sibling: someone who knows more, will say so, will guide, and is not scoring anyone. This distinction is not cosmetic. A teacher's authority is positional — it arrives with the role and it is precisely what attaches a verdict to not-knowing. The sibling relation is asymmetric in knowledge without being evaluative, which permits guidance to be given and received without the transaction implying judgement.

Naming this aloud mattered. Participants arrive with a strong prior about what a person standing at the front of a room is for, built over a decade of schooling. Stating the relation explicitly is what allowed them to stop running the inherited script — and as Section 5 documents, it still took approximately two sessions before they believed it.

3.3 What we deliberately did not do

Several omissions were deliberate and are recorded here because they are the substance of the experiment. We did not check whether recommended study had been done. We did not intervene when the room went quiet. We did not redirect discussion toward topics we considered more productive. We did not follow up on absences to encourage return. We did not simplify material to preserve engagement, and we did not accelerate it to retain the advanced.

The intent throughout was that any participation observed should be attributable to the participant's own choice, and therefore interpretable.

4. Method and Data

4.1 Cohort

Thirty-five participants registered. Of these, approximately three or four were existing SPARK members; the remainder — roughly 90% of the cohort — were new to the organisation. The majority were in Grades 11 and 12, with the range extending from Grade 8 at the youngest to university undergraduates at the oldest. Participation was free and open; there was no selection process, entrance test, or prerequisite.

The cohort's most consequential property was the extreme heterogeneity of its technical preparation. It included Grade 8 students who had not encountered calculus, university students who had formally studied calculus but only as an examination procedure with no conceptual access to it, participants operating a computer for programming purposes for the first time in their lives, and participants comfortable enough with the material to find the pace insufficiently demanding. Section 7 argues that this single fact is sufficient to explain the technical track's results without reference to the experiment's variables at all.

4.2 Instrumentation

Feedback was collected through voluntary post-session forms after each of the first five sessions, and a longer programme-wide instrument at the conclusion. No form was compulsory, no participant was pursued for a response, and no incentive was attached to completion. Sixty-seven submissions were received in total.

Instrument	Est. attending	Submissions	Rate
Day 1 form	~35	17	~49%
Day 2 form	~17	7	~41%
Day 3 form	~17	6	~35%
Day 4 form	~30	10	~33%
Day 5 form	~30	8	~27%
Final programme form	~30	19	~63%

Table 2. Instrument response rates against estimated attendance. Attendance figures are facilitator estimates; the camp kept no register by design.

Daily instruments captured session rating (1–5), pace, most valuable activity, clarity of each block, likelihood of recommendation, and three open free-text fields. The final instrument added overall growth, self-assessed evolution of understanding, achievement against personal goals, preparedness across four dimensions, and extended free-text on defining moments and proposed structural changes.

4.3 Analytical cautions

Three limitations of the data are severe enough to state before any result is reported, rather than after.

- Absence of attendance records. The camp kept no register, by design. All attendance figures in this report are facilitator estimates and should be treated as approximate. This was a genuine cost of the experimental commitment, and Section 12 recommends resolving it without compromising the design.

- Voluntary response bias. Every instrument was optional, and roughly a third of the final cohort did not complete the closing survey. Participants for whom the model did not work had the least reason to submit feedback and the greatest freedom to simply stop attending.
- The daily instrument did not measure the dependent variable. Its questions concerned satisfaction, clarity and pace. The camp's actual objective was curiosity and self-attribution. The evidence for the primary finding in Section 8 is therefore incidental — it appears in free-text fields designed for other purposes, and survives principally because this cohort wrote at unusual length. Section 12 proposes a corrected instrument.

5. Findings I — Participation and the Adaptation Curve

The most robust result in this study is not an outcome measure. It is the shape of participation over time, which appears in three independent measures and describes the cost of admission to a non-evaluative environment.

5.1 The collapse

Thirty-five participants registered and approximately thirty-five attended the first session. By the second session attendance had fallen to roughly seventeen — a decline of approximately 50%.

The cause was not dissatisfaction. Day 1 ratings averaged 4.24 out of 5 across seventeen respondents, with no rating below 3, and 88% describing the pace as just right. Participants who left were not reporting a bad experience. They were reporting an unintelligible one.

Day 1 feedback records the confusion directly. A discussion block running to three hours with no syllabus and no conclusion was described by one participant as something to be survived. Another objected specifically that the political discussion reached no conclusion — which is an accurate description of open enquiry and a reasonable complaint from someone expecting a lesson. A third asked for the discussion sessions to be restructured toward questions students had about their studies rather than topics that mainly lead to debate.

> *I think the discussion sessions could be improved by focusing on topics that are more helpful and relevant to students. It would be great to ask students what questions or confusions they have about their studies and discuss those, rather than choosing topics that mainly lead to debates.*
>
> — Day 1 feedback, anonymous

This is the sound of a student attempting to map an unfamiliar room onto the only model of a classroom they possess. The request is entirely sensible under that model and entirely at odds with the design. Roughly half the cohort resolved the mismatch by not returning.

5.2 The recovery

Attendance recovered to approximately thirty by the fourth session — around 86% of registration — and held at that level through to the end. Critically, the absences recorded from that point forward became meaningful: participants pre-informed facilitators of reasons, which had not occurred during the trough.

Session	Est. attending	Mean rating	Pace "just right"	Mean recommendation
Day 1	~35	4.24	88%	not collected
Day 2	~17	4.14	86%	4.14
Day 3	~17	4.67	100%	4.83
Day 4	~30	4.50	90%	4.50
Day 5	~30	4.62	88%	4.75
Final	~30	4.00 (growth)	—	4.68

Table 3. Participation and satisfaction by session. Ratings on a 1–5 scale.

The significance of this table is easily missed. Session ratings held steady — and in fact rose — across precisely the window in which the cohort nearly doubled by re-absorbing its most sceptical members.

Had the recovery consisted of a loyal core, a rising mean would be unremarkable survivorship. It did not. Returning the disoriented to the room did not depress the measure.

We take this as evidence that the Day 1 problem was one of legibility rather than fit. Students did not leave because the environment was wrong for them; they left because they could not yet read it.

> **The adaptation cost, quantified**
>
> Approximately two sessions and approximately 50% temporary attrition is the price of transition into organised freedom of thought. This is a design parameter, not a failure — but it carries a hard operational implication: the model cannot function in engagements shorter than about four sessions. Anything of three sessions or fewer delivers only the trough.

5.3 Voice followed the same curve

A third measure moved in the same shape. Participants were, by facilitator observation, notably reticent through the early and middle sessions, and became substantially more vocal only in the final sessions. The Day 5 request that presentations be made voluntary rather than universal, and the marked increase in length and candour of final-survey free-text responses, are both consistent with a cohort operating comfortably for the first time.

Attendance, voluntary form submission and willingness to speak therefore describe the same J-curve with the same inflection point. Three independent measures converging on one shape is the strongest quantitative claim this study can make: the trough is a property of the model, not an artefact of any single metric or any one cohort's mood.

A consequence worth stating plainly is that this six-session programme contained roughly two sessions of adaptation and four sessions of the intended environment. We have therefore never observed the model at steady state — only its approach to it.

An incidental predictor

Of the twelve participants who voluntarily gave their names on the Day 1 form, nine appear in the final survey — approximately 75%, against a cohort in which roughly half the room disappeared for two weeks. Voluntarily completing an optional form on the first day was, in this cohort, a strong early signal of eventual retention. Section 12 proposes using this operationally: not to enforce anything, but to identify within twenty-four hours who is likely to be in the trough, so that they can be reached before the third week decides for them.

5.4 A second, accidental replication

The J-curve was run twice. The camp was publicly announced and then delayed by approximately two weeks before commencing. Asked afterwards what they had been thinking during that delay, one participant offered a two-part answer which describes the identical shape occurring before anyone had attended a single session:

> *ehh idk abt this summer camp anymore ... yeah nvm it was actually SO worth it in the end tho*
> — Participant, recalling the pre-camp delay

Negative priming before first contact, fully recovered by conclusion. Two independent shocks — one administrative, one pedagogical — and two complete recoveries. We did not design this and would not have thought to. It is nonetheless the closest thing in the dataset to a robustness check.

5.5 Retention in context

Approximately thirty of thirty-five participants completed the programme: roughly 86% terminal retention. The conditions under which this figure was achieved deserve emphasis. The camp was free, so no financial sunk cost held anyone. Attendance was unrecorded, so no institutional consequence held anyone. Approximately 90% of the cohort were new to SPARK, so no prior loyalty held anyone. There was no certificate, no grade, and no credential at stake.

Free programmes typically bleed participants precisely because nothing holds them. In this case nothing held them and they returned regardless.

Additionally, a number of students who never registered — many having assumed, because of university admissions timelines, that they could not commit — remained in SPARK's community channel throughout and expressed regret at having missed the programme. This group has no sunk cost and no social obligation to perform enthusiasm; their response was generated entirely by secondhand accounts from peers. It also identifies the actual leak in the funnel: participants self-selected out on a projected-availability judgement before ever encountering the product, when partial attendance had always been perfectly acceptable.

6. Findings II — The Discussion Track

The discussion block carried the programme. This is not a subjective impression; it is the most consistently replicated result in the dataset.

6.1 Measured dominance

Measure	Discussion block	Technical block
Selected as most enjoyable activity (all sessions)	37 selections	21 selections
Rated "Unclear" (Days 2–5)	1 response	4 responses
Named as most valuable to personal development	14 of 19	9 of 19
Dominant theme in "most fun experience" free-text	overwhelming majority	isolated mentions

Table 4. Comparative standing of the two blocks. Multiple selections permitted.

In the final survey, participants selected which aspects of the camp had been most valuable to their personal development. The ranking places the human and dialogic elements above the curricular ones:

Aspect	Selections (of 19)
Mentorship and feedback	15
Debates and discussion	14
Networking with peers	12
Hands-on coding	9
The pace and intensity of the curriculum	7

Table 5. Most valuable aspects, final survey. Multiple selection.

6.2 What actually happened in the room

The discussion block had no syllabus, and its subject matter across six sessions was determined entirely by what participants raised. The recorded range includes abiogenesis, theology, epistemology, the historical formation of mythologies, vampire folklore, geography, cinematography, art, literature, physics, gender and social structure, the ethics of research internships in Pakistani universities, and career direction.

The characteristic structure was not topical but associative — one subject producing the next. One participant described a single evening moving from a peer's presentation on shadows into magic, into the concept of angels and devils, and into the history of vampires, and identified the movement itself as the thing she valued:

> *I enjoyed how the conversation moved naturally from one topic to another, and everyone shared their own perspective. It was interesting because I learned new things while also getting to hear different points of view.*
>
> — Final survey, university-level participant

The same participant, asked what topics she wanted next, explicitly requested that no fixed topics be set — on the grounds that the natural drift was the mechanism. That a participant independently identified and defended the design principle is the clearest available evidence that the model had become legible to the cohort by the end.

6.3 The real output: appetite, not knowledge

The discussion block was not an efficient knowledge-transfer mechanism and was not intended as one. Its measurable output was appetite — specifically, self-generated questions about material nobody had taught.

The evidence sits in the final survey's field asking what participants wished to explore next. The answers are not requests for topics already covered. They are questions the participants had constructed for themselves, adjacent to things the room had merely brushed against:

- One participant, rather than naming a subject, posed a research question: how ideological groups containing serious internal disagreement can be united behind a single cause, and what magnitude a cause must reach to make that possible.
- Another submitted a list including the simulation hypothesis, the hard problem of consciousness, the Ship of Theseus, quantum entanglement, the arrow of time, Gödel's incompleteness theorems, entropy, free will against neuroscience, mathematical Platonism, and why there is something rather than nothing — noting that these sit at the intersection of science, philosophy, mathematics and theology.
- Another requested the six-planet parade and the Perseid meteor shower, having encountered them in the news, and framed the request as wanting to discover whether the interest would develop.
- Others named Mohenjo-daro and lost civilisations, Islamic history studied slowly enough to be lived rather than memorised, cybersecurity, robotics, game development, and creative writing.

These are Grade 11 and 12 students in Pakistan, weeks away from a board-examination system in which questions without marks attached have no function. That they left writing down questions with no examination behind them is, in our assessment, the single most important observation in this report.

A second signature is behaviour outside the room. One participant reported that she had begun reading about historical events between sessions, unprompted, and noted that she had never previously considered herself someone interested in history. Curiosity that survives contact with a person's own self-concept is a stronger result than curiosity expressed inside a warm room at the end of a good evening.

6.4 The cost of moderation-only

The discussion block is simultaneously the programme's strongest asset and the source of its only observed failure mode. Two incidents are recorded in the data.

On Day 3, a participant described an exchange in which she felt she had unintentionally given the impression of arguing when she was in fact seeking clarification, and separately described the difficulty of being expected to immediately suppress a sudden rush of emotion. On Day 4, a discussion of feminism and gender produced simultaneously the session's most positively-cited moment and its most negatively-cited one — one participant naming the balanced treatment of two conceptions of feminism as the day's best element, another recording the same discussion as an unwanted war.

The final survey contains two independent reflections on this:

> *I noticed some people's lack of control on their words and actions (I am no exception ofc) so maybe if it stayed to being just a debate rather than a personal argument.*
> — Final survey, Grade 12 participant

Our reading is that this is not an argument for softer topics. The difficult topics are precisely what participants valued and precisely what produced the perspective shifts documented in Section 8. It is an argument for explicit norms of argument — steelmanning, separation of position from person, a designated moderating role — established by group consent at the first session.

This costs nothing on the free-will axis. Norms adopted by the room are the participants' own rules rather than an imposed authority, which is exactly the Lyceum arrangement: hierarchy-free, but not norm-free. It closes the only failure mode the data actually exhibits.

7. Findings III — The Technical Track

The technical block produced high engagement, genuine individual breakthroughs, and low aggregate skill transfer. This section argues that the last of these is fully explained by two structural factors and carries no information about the experiment's hypothesis.

7.1 The measured result

Preparedness dimension	Very prepared	Somewhat	Not prepared
Confidence in applying knowledge	12	6	1
Problem-solving ability	9	9	0
Collaboration skills	8	10	1
Technical skills	3	15	1

Table 6. Self-assessed preparedness, final survey (n=19).

Technical skill is the flattest line on the instrument. Confidence and problem-solving both substantially outperform it. Additionally, fourteen of nineteen participants reported having only partially achieved the personal goals they set at the outset, against three who achieved them and two who exceeded them — while simultaneously returning a mean recommendation score of 4.68, with fourteen of nineteen awarding the maximum.

That combination — very high satisfaction alongside moderate goal achievement — is the central tension in the quantitative data, and it resolves cleanly once the two structural causes are identified.

7.2 Cause one: irreducible heterogeneity

The cohort contained, within a single room and a single curriculum, at minimum:

- A Grade 8 participant with no exposure to calculus.
- University students who had formally passed calculus courses but possessed no conceptual access to the subject beyond examination technique.
- Participants operating a computer for programming purposes for the first time in their lives.
- Participants advanced enough to find the pace insufficient, one of whom remarked on the general absence of calculus knowledge in the room and requested advanced material on the explicit understanding that everyone else would suffer.

These are not points on a single spectrum. They are distinct problems that happen to produce identical wrong answers. A student who lacks the machinery and a student whose machinery is welded to an examination context they cannot leave require opposite interventions — and the second is arguably the harder case, because something must be dismantled before anything can be built.

No single-track curriculum survives this spread. Not ours, and not a compulsory one. The flat technical result was therefore overdetermined before the first session and cannot be attributed to the absence of grading or attendance.

> **The same property, opposite signs**
>
> Heterogeneity was a liability for the technical block and an asset for the discussion block. A room containing both a first-time computer user and a student impatient for advanced mathematics is unworkable for cumulative skill acquisition and ideal for a conversation in which stories produce further stories. This — rather than motivation or discipline — is the principal reason the two tracks diverged.

7.3 Cause two: cadence and the nature of cumulative skill

Discussion is effectively stateless. Each session stands alone; nothing is lost between meetings; a participant who misses a week has missed one conversation, not a prerequisite. Programming is cumulative. It requires rehearsal between exposures or it does not consolidate, regardless of how engaging the exposure was.

With a weekly cadence and no compulsory practice, the technical block delivered approximately six isolated exposures to a subject that requires repetition. Day 5 feedback records the predictable consequence:

> *Today's lecture was far difficult, just flying over the head, was not able to understand how to code.*
> — Day 5 feedback, anonymous

It is important to note what this does and does not test. A compulsory camp meeting weekly with no practice would produce the same flat result — and would additionally produce resentment. The honest conclusion is that the hypothesis remains untested on cumulative skill acquisition, because the study contains a single arm and no control.

Why the preparation model worked for one track and not the other

Participants received a short message before each session indicating what to study, framed so that they would arrive with questions rather than with completed work. This is a deliberate inversion: preparation ceases to be an obligation owed to a teacher and becomes a precondition for extracting personal value from the room. The behaviour is identical; the incentive structure is inverted; no coercion is required.

It also explains the divergence precisely. Discussion converts arriving questions directly into value. Programming requires rehearsal, and "arrive with a question" does not generate rehearsal. The flat technical outcome is therefore evidence that the question-generation model is domain-specific — not evidence that voluntary learning fails.

7.4 What the technical track did achieve

Aggregate skill transfer was low. Individual outcomes were not, and the programme's stated objective was never expertise. A six-session camp cannot produce machine-learning engineers; neither, reliably, can a four-year degree costing a substantial fraction of a family's savings.

What the technical block did produce:

- A Grade 8 participant who learned calculus, fitted a model and generated predictions — capability essentially absent from her age cohort, and more significantly a counterexample she now owns regarding what she is permitted to attempt.
- Participants who understood, frequently for the first time, that artificial intelligence is a substantially larger field than large language models, and that web development involves considerably more than design and code.
- A university participant experiencing difficulty in her degree who reported gaining multiple perspectives on the subject rather than the single examination-oriented one available to her.
- First working programs. One participant recorded the day's best moment as finally getting her code to run without errors; another, having coded for the first time in her life, noted it had been six or seven years since her last formal computer class.
- Directional clarity. At least one participant identified web development as a career direction and stated an intention to build a full plan around it.

The correct framing is that the technical block functioned as a high-quality exposure surface rather than a training programme, and should be designed and evaluated as such.

8. Findings IV — The Central Mechanism

The principal finding of this study is not a score. It is a mechanism, visible in multiple independent cases, which we believe accounts for the whole pattern of results.

> **The finding**
>
> The burden of formal education is not difficulty. It is the verdict attached to not-knowing. Remove the verdict, and difficulty ceases to produce avoidance — it produces re-attribution and persistence instead.

8.1 The attribution signature

The cleanest evidence in the dataset is not in the rating scales. It is in the pattern of how participants explained their own difficulty. In a graded environment, difficulty converts into either resentment of the institution or damage to the learner's self-concept — the conclusion that one is simply not capable. Neither appears in this data. What appears instead is attribution to method, preparation or circumstance, coupled with a stated intention to continue.

> *Mostly maths, as I've not studied maths in the vacations — that most probably my fault.*
>
> — Day 1 feedback, school-level participant

> *I found the course class a little difficult because I couldn't fully understand what we were trying to do at first. But I'll study and practice it on my own, so I'm sure I'll be able to understand and get it done.*
>
> — Day 5 feedback, university-level participant

> *This is honestly my first time touching something related to computer ... it's hard. It's not that I chose neutral because there's an issue with teaching. It's just that I'm completely new to it, but as I find it interesting, I'll work on improving it.*
>
> — Day 4 feedback, participant new to computing

In each case the participant encountered material beyond their current reach, took responsibility without taking damage, and declared an intention to continue. This is the behaviour the hypothesis predicts, and it recurs across participants of markedly different ages and backgrounds.

8.2 The clearest case: re-attribution of a prior failure

One participant had attempted to teach herself machine learning approximately a year before the camp, had failed to understand much of it, and had encoded that failure as evidence about herself. Her final-survey response records the reversal:

> *My biggest lightbulb moment was realising that I don't actually dislike coding — I just never had enough practice or the right environment to understand it. I also realised that when I tried self-studying ML a year ago and couldn't understand much, it wasn't necessarily because I was "bad" at it; I simply hadn't found a way of learning it that worked for me. After these six weeks, I've started to see ML a little differently, and I might actually explore it further, because maybe I'm not as bad at it as I thought.*
>
> — Final survey, university-level participant

The camp did not teach this participant machine learning. It reclassified a year-old verdict from a statement about her capacity into a statement about method. On the account advanced in Section 2, that reclassification is worth more than fluency in any particular library, because it removes the thing that stops people asking questions in the first place.

8.3 The reversal observed mid-dataset

The most complete evidence is a single participant's trajectory across the feedback forms, because it captures the school-conditioned response firing and then failing to hold.

On Day 2, having been encouraged to ask a question, she attempted it and experienced it as a public failure:

> *The mess I created today. That's why I don't ask questions. But you advising to try, I tried to get out of the shell but left embarrassed. Won't do it again. Really sorry.*
>
> — Day 2 feedback

This is precisely the mechanism Sagan describes, captured at the moment of formation: one exposure, one perceived humiliation, and a stated resolution never to ask again.

By Day 4 the same participant was among the most engaged in the room, recording that the session's only flaw was that it ended too soon. By the final survey she was requesting individual career guidance, asking to join the SPARK team, describing a career direction she had identified for herself, and expressing regret that the programme had concluded.

The resolution did not hold. It did not hold because the environment offered no evidence to sustain it — nobody scored the failed question, nobody recorded it, and the following session proceeded as though it had not happened. We regard this trajectory as the single most informative case in the dataset, and note that no conventional evaluation instrument would have detected it.

8.4 A parallel case: difficulty without shame

A second participant, new to the field, reported no single defining moment and instead offered a realisation about the nature of the subject itself:

> *I don't think I had one specific "lightbulb" moment, as I was completely new to this field and was mostly trying to understand the basics. But I did have one important realisation: now I know that coding is not as easy as it looks! Before SPARK, I had no idea how challenging it could be to learn coding from scratch. So, in a way, realising that was my own little lightbulb moment.*
>
> — Final survey, participant new to computing

The same participant asked, in the topics field, to begin again from the absolute basics — describing herself as being at roughly a Grade 4 level in coding, and doing so cheerfully. A student who can request elementary material by name, in writing, without embarrassment has ceased to treat not-knowing as a verdict. That is the outcome, stated as plainly as the data states it anywhere.

8.5 Perspective shift as a distinct outcome

Asked to describe the evolution of their understanding, participants divided as follows:

Self-described evolution	Responses (n=19)
I had some knowledge, now I have a deep mastery	6
My perspective shifted significantly	6
I have a much clearer application of these skills	4
I was completely lost, now I am confident	3

Table 7. Self-assessed evolution of understanding, final survey.

Free-text responses make the character of the shift concrete. One participant described the feminism discussion as having exposed her to perspectives she had not previously considered and having changed her position, and connected this explicitly to the technical work — noting that the camp had been about questioning her existing ideas as much as acquiring concepts. Another identified a discussion of the exploitation of student research interns in universities as the defining moment of six weeks, and reported changed behaviour outside the room as a result.

A third articulated the general form of the outcome without prompting:

> *It was realising that we really can do so much more than we think we're capable of — if we have the right aim and are willing to put in the effort, stay consistent and actually work towards it the right way, so much becomes possible.*
> — Final survey, university-level participant

8.6 The facilitator confound

One result must be reported against our own interest. Mentorship and feedback was the single most-selected valuable aspect, chosen by fifteen of nineteen participants — ahead of discussion, peers and curriculum. Free-text responses attribute the environment substantially to the specific people running it:

> *The happiness of being taught by someone who doesn't feel like a typical teacher in a typical classroom environment. Our mentor laughs at the students' silly jokes and participates in their conversations without getting frustrated or losing patience. It made the whole learning experience feel so much more comfortable and enjoyable.*
> — Final survey, school-level participant

This creates a genuine interpretive problem. If organised freedom of thought produces these outcomes only when the facilitators happen to be two people who find abiogenesis and vampire mythology worth discussing late into the evening, then what has been demonstrated is not yet a transferable method.

We note two mitigating observations. First, the facilitator effect is at least partly structural rather than personal: what participants describe is the absence of the evaluative posture, which is a design property and can be adopted by anyone. Second, the facilitators' relative youth appears to have accelerated acceptance — a claim of non-evaluation from a peer-adjacent figure is tested for weeks rather than months. But the confound is real, and until the model is run by someone else it and its originators remain a single variable.

9. Findings V — What Students Requested When Nothing Was Imposed

Subsidiary question three asked what structure participants would request for themselves under conditions of genuine free choice. We regarded this as the most informative question available, because a constraint chosen is not a burden, and because what learners ask for when nothing is imposed measures need directly.

The answer was unambiguous and, to us, unexpected. Given complete freedom from homework, attendance and assessment, participants independently requested structure — and converged on substantially the same proposals without conferring.

9.1 The convergent requests

4. Practice between sessions. Multiple participants requested assignments after each session, explicitly for practice rather than assessment.
5. Pre-session grounding. One participant proposed short pre-learning resources before theory — a beginner-friendly video, a short reading, or a few real-life examples — to supply context for students without background, noting that the theory had been difficult at first and had become clear only as the camp progressed.
6. A review block. One participant submitted a complete four-part session architecture, unprompted: an opening hour in which students recount the previous session and the facilitator gives a light recap so that an absent student can follow, followed by open questions; then theory; then hands-on coding; then a work-checking and debugging block; then a closing period of non-academic open conversation.
7. Theory before code, rather than code before theory.
8. Greater frequency. Requests to increase from weekly to multiple sessions per week were repeated across the dataset, including one participant offering to attend five days a week and to extend the programme by months, and another describing the wait between Wednesdays with impatience.
9. More applied projects and real-world examples, and voluntary rather than universal presentations.

> *Only one day a week honestly felt like it wasn't enough. It was so much fun that I'd actually impatiently wait for Wednesdays every week.*
>
> — Final survey, university-level participant

9.2 Interpretation

The naive reading — that students require imposed structure after all — is wrong, and inverts the result. Nobody imposed these requests. Participants who were free to demand less demanded more, and specified the mechanism by which it should be delivered.

The correct reading is that under genuine free will, learners converge on requesting the scaffolding they actually need. A constraint one chooses does not function as a burden. This is a stronger and more useful claim than the hypothesis we set out to test, and the design implication is precise:

> **Offer, never assign**
>
> Publish optional practice and let uptake be the measurement. If students take it voluntarily, the finding is real and the scaffolding is free of psychological cost. Uptake rate under zero enforcement is itself the experiment — and is the single most valuable instrument the next cohort could carry.

A caution attaches. The present study infers willingness from testimony rather than behaviour. Participants stated that they wanted practice; we do not know how many would have completed it. Section 12 treats this as the priority measurement for the next cohort.

9.3 Demand for future content

Requested topics divided cleanly into two clusters, which is itself informative:

Technical requests	Discussion requests
Coding from absolute basics (most frequent)	Philosophy of physics; consciousness; time
Real-world and applied projects	History; ancient civilisations; culture
Game development; robotics	Theology and religious history, paced slowly
Cybersecurity	Astronomy and current celestial events
AI agents, automation, and monetisation	Creative writing and fiction
Data visualisation tooling	Career direction and self-development

Table 8. Requested future content, final survey.

The technical requests are almost uniformly remedial or applied. Nobody asked for greater theoretical depth. This is entirely consistent with the preparedness data in Section 7 and constitutes independent confirmation of it.

10. Limitations

This study has real weaknesses, and they are stated here in full because a case study which conceals them is worthless to the schools it is addressed to.

10.1 The dataset systematically excludes its own counterexamples

Approximately five participants never returned. Roughly a third of those who completed the programme did not fill in the final survey. Any participant for whom unstructured freedom was genuinely aversive — who needed the scaffold, felt adrift without it, and quietly left — is invisible in every file underlying this report.

Every conclusion here is therefore conditioned on the phrase among those who adapted. The broad hypothesis as originally stated cannot be answered from a sample which structurally discards its own disconfirming cases. Three or four short conversations with non-returners would be worth more than another full cohort measured the same way.

10.2 Selection

The programme was free, voluntary, located near a major university campus, and attracted participants who self-selected into a six-week intellectual camp during their holidays. This cohort is unusually predisposed to succeed without compulsion.

What has been demonstrated is that organised freedom works for students who will opt into it. That remains a useful and non-trivial claim. It is narrower than the proposition that the burden of schooling is unnecessary in general, and the distance between those two claims is where most education-reform arguments fail.

10.3 No control arm

The study has a single arm. The flat technical result is consistent with two incompatible explanations — that voluntarism fails to generate sufficient time-on-task for cumulative skills, or that a weekly cadence fails to do so regardless of compulsion — and this design cannot distinguish them.

10.4 The instrument measured the wrong variable

As stated in Section 4.3, the forms asked about satisfaction, clarity, pace and preparedness. The dependent variable was curiosity and self-attribution. The evidence for the central finding is incidental, surviving only because this cohort wrote at length in free-text fields intended for other purposes.

10.5 Self-report and immediacy

Perspective shift is self-reported, and reported in a warm room at the conclusion of an experience participants enjoyed. Curiosity restoration is meaningful only if it persists beyond the social context that produced it. The honest test is a follow-up at six months — not a survey, but a single question in the community channel: what have you learned since, that we did not teach you?

10.6 Calendar compression

The programme was framed and marketed as a six-week camp. Following the initial delay, six sessions were delivered across approximately three and a half weeks, with the final two weeks running consecutive-day sessions. The label and the delivery do not match, and the compression reduced the between-session interval precisely where cumulative technical material most needed it.

11. Design Principles

The following are stated as transferable principles for any educator attempting to reproduce this model. They are ordered by how much they mattered.

1. Remove the verdict before removing anything else

The operative variable is not homework, attendance or grading in themselves. It is whether anyone in the room holds the power to certify a student as incapable. Everything else follows from removing that, and removing the other apparatus without removing the verdict achieves nothing.

2. Expect the trough and hold your nerve

Attendance, participation and voice will collapse for roughly two sessions before recovering. This is the model working, not failing. An educator who does not expect it will read week two as a verdict on the programme and abandon it approximately one session before the recovery begins. If a single fact is transmitted to anyone attempting this, it should be this one.

3. Do not attempt this in fewer than four sessions

A programme of three sessions or fewer delivers the adaptation cost without the payoff. Single-day workshops in this format are not advisable and should be designed differently.

4. Invert the purpose of preparation

Tell participants what to study so that they arrive with a question, not so that they arrive having complied. Do not check. The behaviour is identical; the psychology is opposite.

5. Separate stateless from cumulative content

Discussion tolerates heterogeneity, irregular attendance and weekly cadence. Programming tolerates none of these. The two want opposite population designs and should not be forced to share a single track or a single evaluation standard.

6. Norms are not hierarchy

Establish rules of argument by group consent at the first session. The Lyceum was hierarchy-free, not norm-free. This costs nothing on the freedom axis, because the rules belong to the room, and it closes the only observed failure mode.

7. Name the relationship out loud

Students arrive with a decade-old prior about what a person at the front of a room is for. Stating explicitly that you are not a teacher and are not scoring anyone is what permits them to stop running the inherited script — and they will still take about two sessions to believe it.

8. Offer scaffolding; never assign it

Learners given real freedom will ask for structure. Provide it as an option and measure uptake. Uptake under zero enforcement is the only honest evidence that the structure was wanted rather than tolerated.

9. Remain available outside the room

The highest-value outputs of this camp were individual: career direction, personal guidance, a participant's decision about a degree. These arrived through a community channel and a feedback form, which means they depended on the student being willing to ask twice. Some will not ask at all. A standing, low-ceremony offer — an hour a week, open to anyone, no reason required — surfaces the participants who most need it and are least likely to request it.

12. Implications for Partner Schools

This section states what we believe follows for institutions, and is deliberately conservative about what the evidence supports.

12.1 What we are not claiming

We are not claiming that schools should abolish assessment. Schools are obliged to certify, certification requires measurement, and no evidence in this report bears on whether that machinery can be dispensed with. A programme with 35 self-selected volunteers and no credential at stake cannot speak to the operation of a school.

We are also not claiming that this model produces technical competence. It did not, at this cadence and with this heterogeneity, and Section 7 explains why.

12.2 What we are claiming

10. The burden students carry is attached to the verdict, not the difficulty. Difficult material presented without evaluative consequence did not produce avoidance in this cohort. It produced persistence and self-attribution to method. Any environment a school can create in which not-knowing carries no consequence should be expected to produce some of this effect.
11. Curiosity in Grades 11 and 12 is suppressed rather than absent, and it is recoverable within weeks. This cohort left generating unprompted questions across philosophy, physics, history and theology, and in at least one documented case took up independent reading in a subject the participant had not believed interested her.
12. A non-evaluative space is complementary to schooling, not a substitute for it. It occupies precisely the ground that a certifying institution structurally cannot, which is why partnership rather than replacement is the appropriate relationship.
13. The transition has a measurable, budgetable cost. Roughly two sessions of disorientation and a temporary halving of participation. An institution that expects this will not misread it as failure.

12.3 What a partnership can look like

- A recurring non-evaluative discussion period — moderated, unsyllabused, explicitly ungraded — of at least four consecutive sessions.
- Facilitator briefing on the trough, on moderation without lecturing, and on norms of argument established by group consent.
- Technical exposure sessions framed and evaluated as exposure surfaces rather than training, streamed by prior preparation where numbers allow.
- A standing individual-guidance offer alongside group sessions, since the highest-value outcomes in this study were individual.
- Instrumentation as proposed in Section 13, so that the institution measures curiosity rather than satisfaction.

13. Open Questions and the Next Cohort

The following changes would convert a strong qualitative result into a defensible one.

13.1 Measure the actual dependent variable

Three questions would achieve more than the entire instrument used in 2026:

- What did you look up on your own this week that nobody asked you to?
- What is a question you have now that you did not have six weeks ago?
- Describe a moment when you did not understand something. What did you do, and how did it feel?

The third probes the re-attribution mechanism directly, and is the question this study most wishes it had asked.

13.2 Make uptake the experiment

Publish optional practice material and measure completion under zero enforcement. This converts the Section 9 finding from testimony into behaviour, and is the single highest-value change available.

13.3 Instrument attendance without enforcing it

Record who attends; require nothing, pursue no one, attach no consequence. The 2026 cohort's mid-camp visibility was lost entirely, and it remains impossible to determine whether the low form response during the trough reflected disengagement or simple form fatigue. Recording is not enforcement.

13.4 Resolve the cadence confound

Run at the frequency participants requested — two to three sessions weekly, with voluntarism entirely intact. If cumulative skill still fails to form, that is a genuine negative result about voluntary learning. If it forms, cadence is isolated as the variable and the hypothesis survives intact. Either outcome is worth more than the ambiguity currently held.

13.5 Stream the technical track

Separate by prior preparation. This is matching, not compulsion, and costs nothing on the freedom axis. The 2026 cohort's most advanced participants were the least well served, by their own accounts.

13.6 Follow up at six months

A single question in the community channel. Durability is the difference between having demonstrated a lasting effect and having run a very good six weeks — the latter being worthwhile, but a smaller claim.

13.7 Test transferability

Participants have been asked to carry the model outward: to learn, to spread it, and to become capable of standing in front of a room and telling others to conquer themselves and the world, and to leave it better for those who follow rather than worse.

Two cautions attach to that instruction, both drawn from this data. First, what transmits easily is the aesthetic — it was free, we discussed everything, nobody was mocked — while what actually does the work is the discipline underneath: moderation without lecturing, the nerve to hold through the trough, preparation framed as question-generation, and the restraint not to lecture at a confused student. A facilitator who runs on enthusiasm alone will meet the trough in week two, read it as failure, and stop. If one thing is transmitted, it should be the J-curve.

Second, the participants most moved by this camp were those who arrived with the least technical preparation, while the most technically capable reported gaining the least. The multiplication vector therefore runs through people equipped to transmit the discussion model rather than the technical one. This is acceptable if the discussion model is the product — but it should be a deliberate choice rather than a discovery.

The measurable form of this objective, and the only one that distinguishes a camp from a method, is replication count: how many of the thirty run something of their own within a year. Not attendance, not ratings. Replication.

14. Conclusion

We set out to test whether the psychological burden of schooling is produced by difficulty or by evaluation. The camp removed evaluation entirely and kept the difficulty — calculus for Grade 8 students, machine learning for participants who had never programmed, and open argument on theology, gender and history with no answer key at the end.

Half the room left within two weeks, because a classroom with no verdict is unreadable to students who have never encountered one. Then they came back, and stayed, and by the end they were asking for more of it than we were able to provide.

The programme did not produce engineers, and was not designed to. What it produced is documented in the words of the participants themselves: a Grade 8 student who fitted a model and now knows what she is permitted to attempt; a university student who stopped believing she was bad at machine learning; a participant who resolved on Day 2 never to ask a question again and by the final session was asking for career guidance and to join the team; a student who began reading history for no reason at all; and a set of questions about consciousness, time, incompleteness and lost civilisations that nobody assigned, that will never be examined, and that were written down anyway.

Sagan's observation was that something terrible happens between kindergarten and high school. Our finding is narrower than a solution and more useful than a lament: what happens is that not-knowing acquires a verdict, and questions become expensive to ask. In a room where that price is not charged, and given about two sessions to believe it, sixteen-year-olds start asking again.

That is the whole of the result. We think it is enough to build on.

Appendix A — Instruments

Post-session form (Days 1–5)

- How would you rate your experience today? (1–5)
- What was the best part of today? (free text)
- What was one thing you did not enjoy or found difficult? (free text)
- How did you feel about the pace of the activities today? (too slow / just right / too fast)
- Which activity did you find most fun today? (course class / discussion / both)
- Do you have any suggestions to make the next session better? (free text)
- How likely are you to recommend the camp to a friend? (1–5) [added from Day 2]
- Clarity of instructions and topics — course class and discussion, separately (unclear / neutral / very clear) [added from Day 2]
- Which topic or skill area would you like to explore next? (free text) [added from Day 2]

Programme-wide form

- Overall growth and learning across the camp (1–5)
- Evolution of understanding from start to finish (four options)
- Extent of achievement against personal goals (four options)
- Most defining or lightbulb moment (free text)
- Most valuable aspects to personal development (multi-select)
- One change to structure or content (free text)
- Preparedness across technical skills, problem-solving, confidence, collaboration
- Most fun experience (free text)
- Likelihood of recommendation (1–5)
- Topics for future sessions (free text)

Appendix B — Summary Data

Metric	Value
Registered participants	35
Estimated terminal retention	~30 (~86%)
New to SPARK	~90% of cohort
Sessions delivered	6
Total feedback submissions	67
Trough attendance (Days 2–3)	~17 (~50%)
Mean daily session rating (Days 1–5)	4.24 / 4.14 / 4.67 / 4.50 / 4.62
Final recommendation score (mean)	4.68 of 5
Maximum recommendation scores	14 of 19
Final overall growth (mean)	4.00 of 5

Metric	Value
Goals partially achieved	14 of 19
Technical skills — very prepared	3 of 19
Confidence in applying knowledge — very prepared	12 of 19
Most valuable aspect (top)	Mentorship and feedback (15 of 19)
Discussion vs course class — enjoyment selections	37 vs 21

SPARK Chapter · Islamabad · Summer Camp 2026 Case Study

All participant data has been anonymised. Quotations are reproduced verbatim from voluntary feedback submissions, with minor corrections to spelling and punctuation for readability only; no wording has been altered. This document may be shared and reproduced for educational purposes.